



MK-D-5003 MMA
SDS 01-18
Effective Date 21/02/2023

Safety Data Sheet

1. Identification

Material Name : Methyl 2-methylacrylate, Methyl methacrylate
Synonyms : MMA
CAS Number : 80-62-6
Product Use : Raw material for many types of polymer products or chemicals.
Manufacturer/Supplier : **Thai MMA Co.,Ltd.**
Address : 1 Siam Cement Road, Bangsue, Bangkok 10800
Section : Supply Chain Department
Telephone number : +66-2586-5875
Fax number : +66-2586-5393
Emergency Telephone number : +66-3891-1714 or +66-3891-1714

2. Hazards Identification

GHS Classification:

Health	Environmental	Physical
Acute Toxicity – Not Classified (oral)	Aquatic Toxicity	Flammable Liquid -
Acute Toxicity – Not Classified (dermal)	Category 3 (Acute)	Category 2
Acute Toxicity - Category 4 (inhalation)	Aquatic Toxicity –	Spontaneous
Eye Corrosion/Irritation - Category 2A	Not Classified	Ignition Liquid –
Skin Corrosion/Irritation - Category 2	(Chronic)	Not Classified
Inhalation Sensitization - Category 1		
Skin Sensitization - Category 1		
Mutagenicity – Not Classified		
Carcinogenicity - Not Classified		
Reproductive/Developmental - Category 2		
Target Organ Toxicity (Single) - Category 3		
(Respiratory Sensitization, Anesthetic activity)		
Target Organ Toxicity (Repeated) - Category 1		
(Respiratory system, Central nervous system)		



MK-D-5003 MMA
SDS 01-18
Effective Date 21/02/2023

Safety Data Sheet

GHS Label:

Symbols:



Hazard Statements

DANGER!

Highly Flammable Liquid and Vapor.

Harmful if inhaled.

Causes skin irritation.

Causes serious eye irritation.

May cause allergy or asthma symptoms or breathing difficulties if inhaled.

May cause allergic skin reaction.

Suspected of damaging fertility or the unborn child.

May cause respiratory irritation.

May cause drowsiness or dizziness.

Causes damage to organs (respiratory apparatus, nervous center) through prolonged or repeated exposure.

Harmful to aquatic life.



MK-D-5003 MMA
SDS 01-18
Effective Date 21/02/2023

Safety Data Sheet

Precautionary Statements

Obtain special instructions before use. Do not handle until all safety precautions have been read and understood.

Wear protective gloves/protective clothing/eye protection/face protection/respiratory protection as required.

Keep away from heat/sparks/open flames/hot surfaces. - No smoking.

Keep container tightly closed.

Ground/bond container and receiving equipment.

Take precautionary measures against static discharge.

Use explosion-proof electrical/ventilating/lighting equipment.

Use only non-sparking tools.

Do not breathe mist/vapours.

Use only outdoors or in a well-ventilated area.

Avoid release to the environment.

Do not eat, drink or smoke when using this product.

Contaminated work clothing should not be allowed out of the workplace.

Wash hands thoroughly after handling.

ORIGINAL CONTROLLED



Safety Data Sheet

Counter measures

IF INHALED: If you feel unwell or breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/physician.

IF ON SKIN: Remove/Take off immediately all contaminated clothing. Wash with plenty of soap and water. If you feel unwell or skin irritation or rash occurs, get medical advice/attention.

IF IN EYES: Rinse cautiously with water for over 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention.

IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician. Do NOT induce vomiting.

IF feeling unwell or experiencing respiratory symptoms: Call a POISON CENTER or doctor/physician.

IF exposed or concerned: Get medical advice/attention.

Collect spillage.

In case of fire, use dry chemical, carbon dioxide, foam, dry sand for extinction.

Storage

Storage in tightly closed container at cool and well-ventilated area.

Store locked up.

Dispose

Contents/container are burned in a designated incinerator or are entrusted to qualified waste trader.

Another Hazards out of GHS classification

This material is very flammable (flash point is 11°C) and vapor may form an explosive mixture with air. (explosion limit is 2.1~12.5vol%)

High temperature or direct rays may cause polymerization. Polymerization followed exothermic heat and may cause explosive reaction.

3. Composition / Information on Ingredients

Component	CAS Number	Weight %
Methyl 2-methylacrylate	80-62-6	99.9%
EINECS Number	TSCA	
201-297-1	Registration	



Safety Data Sheet

4. First Aid Measures

Eye: Eye irritation. Flush immediately with large amounts of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. If possible, remove contact lenses before flushing the eyes. Get immediate medical attention.

Skin: Itching or burning of the skin. Immediately remove contaminated clothing and shoes. Flush the skin with plenty of running water at normal or lukewarm temperatures. Get immediate medical attention if the appearance of the skin changes or if irritation persists.

Inhalation: Remove exposed person from source of exposure to fresh air. Wrap in a blanket or similar material to keep warm. Keep quiet. If breathing is weak or stops, loosen clothing, secure an open respiratory airway and start cardiopulmonary resuscitation (CPR). Avoid mouth-to-mouth resuscitation.

Ingestion: Get immediate medical attention. Do not induce vomiting unless directed by medical personnel. Doing so will increase the potential for harm. Should be allowed to wash the mouths with water, if the person himself wishes. Do not place anything in the mouth if the person himself wishes.

Most important Sign and Symptoms: In case inhalation, may cause dizziness or choking and a nose or the throat irritation. Contact to eye or skin may cause irritation and a burning by chemicals. If swallowed may cause feeling nausea, vomiting or griping.

Protection of rescuer: In case rescue the victims, wear suitable protective equipments.

5. Fire Fighting Measures

Suitable Extinguishing Media: Use dry chemical, foam, carbon dioxide or dry sand to extinguish fire.

Unsuitable Extinguishing Media: Do not use water in a jet. Doing so may be in danger the spread fire.

Special Hazards: Nothing special item.

Fire-fighting procedures: Keep persons not directly involved in fire-fighting out of the area of the fire. Use dry chemicals, a CO₂ extinguisher, dry sand etc. to extinguish a fire in its early stages. In case of large fires, deprive the fire of a source of oxygen using a foam extinguisher or by other means. Wear protective equipment during fire fighting.

6. Accidental Release Measures



Safety Data Sheet

Person protection measures (Protective equipments): Wear protective equipment and operate from upwind during clean up works. Isolate hazard area by rope and deny entry to unnecessary of unprotected personnel.

Environment protection measures: Do not discharge the spill into sewer pipes. If the spill involves small amounts, soak up the material with sawdust, waste cloth or sand, then collect the soaked materials into a container. For extensive spills, block the flow of liquid with earth or sand, cover the liquid surface with foam, then collect as much of the material into an empty container as possible. We recommend using spark proof tools.

Protects from secondary accidents: Promptly remove any flammable materials from the surrounding area. Prepare extinguishing media providing ignition.

7. Handling and Storage

Handling:

- Do not use compressed air to charge, bleed or handle the product.
- If the product is to be handled indoors, provide good ventilation using a local exhaust system or other measures. (In case evaporating at 20 °C atmosphere may be contaminated and the concentration may arrive at poisonous level quickly.)
- Keep the product away from flames, sparks, strong oxidizers and high temperature substances.
- Handle the container gently; do not overturn, drop, strike or drag.
- Wear adequate protective equipment to prevent absorption through the skin, contact with the skin or eyes or inhalation of vapors. -- See Protective equipment.

Storage:

Keep container tightly closed. Store in a well-ventilated place. Keep cool. Store locked up. Keep away from heat, sparks, open flame, hot surfaces - No smoking. Protect from sunlight. Do not store in the same area with organic peroxides.

IMPORTANT: Methacrylates stored in bulk must be kept in contact with air (oxygen). So it will be desirable that the space is sealed by the gas that oxygen concentration is about 8%. This treatment may avoid the space going in flammable limits. The other side for avoiding polymerization, the oxygen concentration of the space needs suitable level. Monomer vapours are uninhibited and may form polymers in vent or flame arresters, resulting in blockage of vents.

Storage Temperature: Store at temperatures not exceeding 35°C. When not using immediately, it is preferable to store at temperature not exceeding 30°C.



MK-D-5003 MMA
SDS 01-18

Effective Date 21/02/2023

Safety Data Sheet

Storage Life: Provided proper storage and handling procedures are followed, the product may be stored for up to 3 months from the date of receipt. If the product is to be placed in long-term storage (over three months), inspect the polymerization inhibitor concentrations for significant declines from initial concentrations as necessary. If polymerization happens and a container heats up, it should be transferred quickly to well ventilated wide area and cool down it by large quantity water.

Recommended Material: Steel drum and Steel 18L can.

8. Exposure Controls / Personal Protection

Control concentration: Not established.

Exposure Limits: **ACGIH**(2011 version) **TLV- TWA:** 50 ppm

ACGIH(2011 version) **TLV- STEL:** 100 ppm

Japanese Society of Occupational Health (2013 version) 2ppm, 8.3mg/m³

Engineering Controls: If the product is used indoors, seal the point of emission or install ventilating equipment or both. Install a safety shower, wash basin and eye bath near the handling area. Indicate the locations of these facilities clearly and prominently.

Personal Protective Equipment (PPE)

Respiratory Protection: Gas mask (to protect against organic gases), airline respirator (in atmospheres with high product vapor concentrations), air respirator, safety goggles or protective shield.

Hand Protection: Avoid skin contact. Wear oil-resistant gloves and boots.

Eye Protection: Wear chemical safety goggles and face shield.

Skin and body Protection: Additional protection may be necessary to prevent skin contact including use of apron, face shield, boots or full body protection.

Occupational Health Measures: Protective Equipment should be inspected periodically. Do not eat, drink or use tobacco when handling this product.

9. Physical and Chemical Properties

Appearance: Clear, colorless liquid

Odor: Ester weak odor

Flashpoint: 11 °C (51.8 °F) (closed cup)

Auto-ignition Temperature: 421°C (789.8°F)

Boiling Point: 100.8°C (213.4°F) @ 1013hPa

Melting Point: -48°C(-54.4 °F)



MK-D-5003 MMA
SDS 01-18
Effective Date 21/02/2023

Safety Data Sheet

Vapor Pressure:	3.73kPa @ 20°C
Vapor Density(Air=1):	3.45(air = 1)
% Solubility in Water:	1.72 @ 20°C
Lower Flammability Limit:	>2.1 vol%
Upper Flammability Limit:	<12.5 vol%
Specific Gravity:	0.944 @ 20°C
pH:	N.A.
Molecular Weight:	100.12
Specific heat:	1.89 J/g·°C
Heat of polymerization:	54.4 kJ/mol
Heat of vaporization:	360kJ/kg
Refractive index(nD²⁰):	1.4152/20 °C
Viscosity:	0.56 mPa·s/20°C
SAPT:	>50 °C

10. Stability and Reactivity

Stability: A polymerization inhibitor is added to the product to stabilize it under normal conditions of handling.

Hazardous Reactivity: Very flammable liquid and the vapor mixed with air generate explosive gas (explosion limit is 2.1~12.5 vol%). Polymerization reaction generate large quantity heat and quick reaction may cause explosion.

Condition to Avoid: Avoid to store far beyond the acceptable period or temperatures.

Incompatibility: Incompatible with strong acids and peroxide. (Cause of polymerization)

Hazardous Decomposition Products: N.A.

11. Toxicological Information

Acute Oral Toxicity:	LD ₅₀ (Rat)	7,900 mg/kg
Acute Dermal Toxicity:	LD ₅₀ (Rabbit)	>5,000 mg/kg
Acute Inhalation Toxicity:	LC ₅₀ (Rat)	7,093 ppm (4H)(Vapor)

Skin Corrosion/Irritation: Skin irritation test by rabbit are reported that moderate irritation is typically observed. PII 1.5 -2.8

Eye Corrosion/Irritation: Eye irritation test by rabbit are reported that the iris and cornea are not impaired. Grade-2 red discoloration is observed after 24hours. Light or middle irritation to eye is observed.



Safety Data Sheet

Respiratory Sensitization: Classified by Occupational Safety and Health Association of Japan as "Respiratory sensitizer 2nd group".

Skin Sensitization: Classified by Occupational Safety and Health Association of Japan as "Skin sensitizer 2nd group".

Mutagenicity: Germ Cell in vivo. Mutagenicity test (dominant lethal) negative.

Carcinogenicity: Classified by IARC as group 3, by ACGIH as group A4, by EPA as class E.

Reproductive and Developmental Toxicity: According to teratogenicity test by rats, the fetal toxicity (early fetal death, decrease of crown-rump length, ecchymoma) is reported at the dosage which develops the maternal toxicity (death, loss of weight, etc.).

Target Organ Effects(Single): Respiratory irritable, cataplexy, favor, nausea, dizziness, headaches and drowsiness are reported based on published human data. So the target organ may be respiratory and central nervous system.

Target Organ Effects(Repeated): Atrophic rhinitis, pharyngitis, damage to autonomic nervous system, nervous depression, headache, dizziness, erethismus, diffuse concentration, failure of one's memory are reported based on published human data. So the target organ may be respiratory system and central nervous system.

12. Ecological Information

Organism Toxicity: Shellfish daphnia magna $EC_{50}/48hr$ 69mg/L.

Persistence/degradability: Readily biodegradable. Degradability by BOD is 94.3%.

Bioaccumulation: Bioaccumulation is not expected to be significant. $\log Kow = 1.38$.

13. Disposal Considerations

Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids.

When incinerating the product, soak up the product using sawdust or similar materials, and burn in small increments. When disposing the products, should entrust to qualified waste trader. We recommend that the Waste Water including these products are treated by activation sludge equipments and are wasted to public space after confirming the quality. When wasting the containers used these products, wash these by water after confirming that the containers do not remain the products.



MK-D-5003 MMA
SDS 01-18
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Safety Data Sheet

14. Transport Information

International Regulations : Follow UN classification

UN Classification : Class3 (Flammable Liquid)

UN Number : 1247

Proper Shipping name : METHYL METHACRYLATE MONOMER, STABILIZED

Packing group : II

Special Safety Measures : Avoid leak the products according to handle the container gently; do not overturn, drop, strike or drag. Watch the temperature of products, especially suddenly temperature raise.

Special Safety Measures and Conditions at Transportation : Follow the instructions given in Handling and Storage (Sec.7). Observe general precautions applicable to flammable toxic liquids. According to Sec.9, SAPT is higher than 50 °C therefore temperature control which is defined at SP386 is not required.

First Aid Measurements Guide Number : 128P

15. Regulatory Information

Note: The latest regulatory information should be confirmed with the importer.

16. Other Information

Revision Indicator: Revised on 17/02/2021

Disclaimer: This SDS was prepared on the basis of materials, data and information available to us at the time of its writing. However, we make no warranty as to the accuracy of the data herein with respect to product content, physical and chemical properties thereof or associated hazards. The precautions given herein apply to normal handling. For handling under special conditions, users are responsible for implementing adequate safety measures.